

Practice Exam #4

SECTION I

Time — 90 minutes
Number of questions — 42
Percent of total grade — 55

1. What is printed when the following code segment is executed?

```
String url = "http://www.usa.gov";  
int pos = url.indexOf("http://");  
if (pos >= 0)  
{  
    System.out.println("<" + url.substring(0, pos) + ">");  
}  
else  
{  
    System.out.println("not found");  
}
```

- (A) <www.usa.gov>
(B) <>
(C) not found
(D) StringIndexOutOfBoundsException
2. Assume that the `int` variables `a` and `b` have been properly declared and initialized. For which of the following pairs of values of `a` and `b` is the value of the following expression true?

`(a > 20 && a < b) || (a > 10 && a > b)`

- (A) 5 and 0
(B) 5 and 10
(C) 15 and 10
(D) 15 and 20

3. Consider the following method.

```
/**
 * Precondition: amt represents a positive value in dollars
 *                  and cents (for example, 1.15 represents
 *                  one dollar and fifteen cents).
 */
private int process(double amt)
{
    return (int) (amt * 100 + 0.5) % 100;
}
```

Which of the following best describes its return value?

- (A) The cent portion in amt
 - (B) The number of whole dollars in amt
 - (C) amt converted into cents
 - (D) amt rounded to the nearest dollar
4. Given two initialized `int` variables `a` and `b`, which of the following expressions evaluates to the same result as $2 * (1 + (3 * b) / a)$ for all positive values of `a` and `b`?
- (A) $2 + 6 * b / a$
 - (B) $2 * (1 + b / a * 3)$
 - (C) $2 + 2 * (b * 3 / a)$
 - (D) None of the above
5. Consider the following code segment.

```
int x = < integer value >;
int y = < another integer value >;

while (x > y && x % y != 0)
{
    x -= y;
}
```

Which of the following conditions will always be true after the `while` loop, regardless of the initial values of `x` and `y`?

- (A) $x < y \parallel x \% y != 0$
- (B) $x <= y \&\& x \% y != 0$
- (C) $x <= y \parallel x \% y == 0$
- (D) $x >= y \&\& x \% y == 0$

6. Consider the following method with a mistake.

```
public static void respondTo(String str)
{
    int fav = str.indexOf("FAVORITE");
    int b = str.indexOf("BOOK");
    int m = str.indexOf("MOVIE");

    if (fav >= 0)
        if (b > fav)
            System.out.println("IT IS MY FAVORITE BOOK, TOO");
        else if (m > fav)
            System.out.println("IT IS MY FAVORITE MOVIE, TOO");
    else if (str.indexOf("NAME") > 0)
        System.out.println("MY NAME IS CHITCHAT");
}
```

It doesn't work as intended, because a pair of braces is missing.

If

```
String str = "THE JUNGLE BOOK MOVIE " +
             "IS BASED ON THE BOOK BY THE SAME NAME.";
```

what is printed when `respondTo(str)` is called?

- (A) Nothing
 - (B) MY NAME IS CHITCHAT
 - (C) IT IS MY FAVORITE BOOK, TOO
 - (D) IT IS MY FAVORITE BOOK, TOO
MY NAME IS CHITCHAT
7. Consider the following statement.

```
System.out.println("\"No\nSmoking\"".indexOf("S"));
```

What is printed when the statement is compiled and executed?

- (A) Nothing, due to a syntax error
- (B) 4
- (C) 5
- (D) 6

8. Consider the following method.

```
private int product(int n)
{
    if (n <= 1)
    {
        return 1;
    }
    else
    {
        return n * product(n-2);
    }
}
```

When you call `product(25)`, it returns `-1181211311`, a negative number. Which of the following best explains this result?

- (A) Integer arithmetic overflow
 - (B) Logic error that shows up for odd values of `n`
 - (C) Stack overflow error in recursive calls
 - (D) Small range of integers in the Java Virtual Machine installed on your computer
9. Consider the following statement.

```
System.out.println(Math.pow(2, 3));
```

What is the result when the statement is compiled/executed?

- (A) 8 is printed
 - (B) 8.0 is printed
 - (C) 9 is printed
 - (D) Syntax error: "pow(double, double) in java.lang.Math cannot be applied to (int, int)"
10. Assume that the boolean variables `a`, `b`, and `c` have been properly declared and initialized. Which of the following expressions is `true` if and only if NOT all three variables `a`, `b`, and `c` have the same value?
- (A) `a != b || b != c`
 - (B) `a != b && b != c`
 - (C) `a > b || b > c || a > c`
 - (D) `!(a == b || b == c || a == c)`

11. Consider the following code segment.

```
double x = 5*4/2 - 5/2*4;  
System.out.println(x);
```

What is printed when the code segment is executed?

- (A) 0
 - (B) 0.0
 - (C) 1.0
 - (D) 2.0
12. Assume that `String` variables `str1` and `str2` have been properly declared and initialized and are not nulls. Which of the following `boolean` expressions evaluate to `true` if and only if `str1` and `str2` hold different values?

- (A) `str1.compareTo(str2) != 0`
- (B) `str1.compareTo(str2) < 0`
- (C) `str1 != str2`
- (D) `str1.equals(str2) != 0`

13. Consider the following code segment.

```
int sum = 0;  
for (int k = 1; k <= 10; k += k)  
{  
    sum += k;  
}  
System.out.println(sum);
```

What is printed when the code segment is executed?

- (A) 10
 - (B) 15
 - (C) 25
 - (D) 55
14. What does API stand for?
- (A) Advanced Program Implementation
 - (B) Actual Parameter Inheritance
 - (C) Application Programming Interface
 - (D) Advance Program Information

15. Consider the following method and the code segment that is part of another method in the same class.

```
public static void process(String word)
{
    word = word.substring(2) + word;
}

String word = "test";
process(word);
System.out.println(word.indexOf("t"));
```

What is printed when the code segment is executed?

- (A) 0
 - (B) 1
 - (C) 2
 - (D) 3
16. Consider the following code segment.

```
for (int r = 1; r < mat.length; r++)
{
    for (int c = 1; c < mat[0].length; c++)
    {
        if ((r + c) % 2 == 0)
        {
            mat[r][c] = 2 * mat[r - 1][c - 1] + c;
        }
    }
}
System.out.println(mat[2][2]);
```

Suppose `mat` is declared as

```
int[][] mat = new int[3][4];
```

and holds the values

```
2 1 3 4
9 7 2 1
0 2 5 6
```

What is printed when the code segment is executed?

- (A) 5
- (B) 11
- (C) 12
- (D) 15

17. Esther wants to write a method `swap` that swaps two integer values. Which of the following three ways of representing the values and corresponding methods will NOT successfully swap the values?

(A)

```
/**
 * a and b are Integer objects that represent
 * the values to be swapped.
 */
public static void swap(Integer a, Integer b)
{
    Integer temp = a;
    a = b;
    b = temp;
}
```

(B)

```
/**
 * a[0] and a[1] contain the values to be swapped.
 */
public static void swap(int[] a)
{
    int temp = a[0];
    a[0] = a[1];
    a[1] = temp;
}
```

(C)

```
/**
 * a[0] and b[0] contain the values to be swapped.
 */
public static void swap(int[] a, int[] b)
{
    int temp = a[0];
    a[0] = b[0];
    b[0] = temp;
}
```

(D) All of the above will work.

18. What is printed when the following statement is executed?

```
System.out.println(17 / 5 % 3 + 17 * 5 % 3);
```

- (A) 1
(B) 4
(C) 9
(D) 42.5

19. Consider the following method.

```
public int countSomething(int[] arr)
{
    int m = arr[0];
    int count = 1;

    for (int k = 1; k < arr.length; k++)
    {
        int a = arr[k];
        if (a > m)
        {
            m = a;
            count = 1;
        }
        else if (m == a)
        {
            count++;
        }
    }

    return count;
}
```

For which of the following arrays will `countSomething` return 3?

- (A) `int[] arr = {0, 1, 1, 1, 1};`
- (B) `int[] arr = {1, 6, 5, 4, 0};`
- (C) `int[] arr = {3, 2, 1, 0, 5};`
- (D) None of the above

20. When a program attempts to open a text file and read data from it, which of the following may cause a run-time error?

- (A) The file is not found in the specified folder.
- (B) The code calls `Scanner's nextLine` method when there are no more lines in the file.
- (C) The code calls `Scanner's nextInt` method when the next token in the file is an integer immediately followed by a comma.
- (D) All of the above

21. What is printed as a result of executing the following code segment?

```
ArrayList<String> digits = new ArrayList<String>();

for (int k = 0; k <= 9; k++)
{
    digits.add("" + k);
}

for (int k = 0; k <= 4; k++)
{
    String d1 = digits.remove(k);
    String d2 = digits.remove(k);
    digits.add(k, d1 + "+" + d2);
}

System.out.println(digits);
```

- (A) [0+1, 1+2, 2+3, 3+4, 4+5]
- (B) [0+1, 2+3, 4+5, 6+7, 8+9]
- (C) [0+1, 1+2, 2+3, 3+4, 4+5, 6, 7, 8, 9]
- (D) [0+0, 1+1, 2+2, 3+3, 4+4, 5, 6, 7, 8, 9]

22. Consider the following code segment.

```
String s = "ban";

ArrayList<String> words = new ArrayList<String>();
words.add(s);
words.add(s.substring(1));
words.add(s.substring(1, 2));

String total = "";

for (String w : words)
{
    total += w;
}

System.out.println(total.indexOf("an"));
```

What is printed when the code segment is executed?

- (A) 1
- (B) 2
- (C) 3
- (D) banana

23. Given `double[][] amps` with 100 rows and 200 columns, which of the following statements assigns to `x` the value in the last row and the first column of `amps`?

- (A) `double x = amps[99][0];`
- (B) `double x = amps[amps[0].length - 1][0];`
- (C) `double x = amps.get(amps.size() - 1, 0);`
- (D) `double x = amps.get(amps.length - 1, 0);`

24. Consider the following class.

```
public class Test
{
    private int t = 0;

    public Test(int x)
    {
        int t = x;
    }

    public int get()
    {
        return t;
    }

    public int increment()
    {
        t++;
        return t;
    }
}
```

What is displayed when the following code segment in a client class of `Test` is executed?

```
Test test = new Test(3);
int x = test.increment();
System.out.println(x + test.get());
```

- (A) 1
- (B) 2
- (C) 3
- (D) 8

25. Consider the following code segment.

```
int[] nums = new int[51];

for (int k = 0; k < nums.length; k++)
{
    nums[k] = 1;
}
for (int k = 3; k <= 50; k += 3)
{
    nums[k] = 0;
}
for (int k = 5; k <= 50; k += 5)
{
    nums[k] = 0;
}
```

How many elements in the array `nums` have the value 0 after this code has been executed?

- (A) 23
- (B) 25
- (C) 26
- (D) 27

26. Consider the following code segment.

```
ArrayList<String> list = new ArrayList<String>();
list.add("A");
list.add("B");
list.add("C");
for (String s : list)
{
    list.set(list.size() - 1, s);
}
```

Which of the following represents the contents of `list` after the code segment has been executed?

- (A) ["C", "B", "A"]
- (B) ["C", "A", "B"]
- (C) ["C", "C", "C"]
- (D) ["A", "B", "B"]

27. Consider the following attempt to open a text file for reading.

```
File textfile = new File("sample.txt"); // Line 1
Scanner sc = new Scanner(textfile);      // Line 2
```

What happens if the file `sample.txt` does not exist?

- (A) An empty file is created and is open for writing.
 - (B) The program continues to run until the first attempt to read from `sc`.
 - (C) The statement on Line 1 throws an `IOException`.
 - (D) The statement on Line 2 throws an `IOException`.
28. Assume that the `String` variables `str1` and `str2` have been properly declared and initialized. Which of the following conditions correctly tests whether the value of `str1` is greater than or equal to the value of `str2` (in lexicographical order)?
- (A) `str1 >= str2`
 - (B) `str1.compareTo(str2) >= 0`
 - (C) `str1.compareTo(str2) == true`
 - (D) `str1.equals(str2) || str1.compareTo(str2) == 1`
29. Suppose sorting an array of 1000 elements using the standard Mergesort algorithm takes on average 40 ms and merging two arrays takes, on average, the number of milliseconds equal to one percent of the sum of their sizes, what is the average time it will take to sort an array of 2000 elements with this algorithm?
- (A) 50 ms
 - (B) 100 ms
 - (C) 160 ms
 - (D) 170 ms

30. Consider the following method.

```
public static String fiboDigits(int n)
{
    int[] fibs = {1, 2, 3, 5, 8, 13, 21, 34};
    String s = "";

    for(int i = fibs.length - 1; i >= 0; i--)
    {
        if (n >= fibs[i])
        {
            s += "1";
            n -= fibs[i];
        }
        else
        {
            s += "0";
        }
    }

    return s;
}
```

Which value does `fiboDigits(30)` return?

- (A) "1010001"
- (B) "1100001"
- (C) "01010001"
- (D) "01100001"

31. What happens when the size of an `ArrayList<Integer> numbers` has reached its capacity and `numbers.add(0)` is called?

- (A) The program throws an `IndexOutOfBoundsException`.
- (B) The first element of `numbers` is set to hold an `Integer` object with the value 0. The size of `numbers` remains unchanged.
- (C) The values in `numbers` are shifted toward the beginning (with the first element overwritten), and an `Integer` object with the value 0 is appended at the end of `numbers`. The size of `numbers` remains unchanged.
- (D) A larger array is allocated to hold the elements of `numbers`; the values from the old `numbers` array are copied into the new array, and an `Integer` object with the value 0 is appended at the end of `numbers`. The size of `numbers` is incremented by 1.

32. Consider the following method.

```
public String fun(String s)
{
    if (s.length() <= 2)
    {
        return s;
    }
    else
    {
        return fun(s.substring(2)) + s.substring(0, 2);
    }
}
```

What is displayed when `System.out.println(fun("ABCDE"))` is called?

- (A) CDEAB
- (B) CDABE
- (C) ECDAB
- (D) DECAB

33. Consider the following class.

```
public class FrequentFlyer
{
    private int miles = 0;

    public FrequentFlyer(int m) { miles = m; }
    public void addMiles(int m) { miles += m; }
    public int getMiles() { return miles; }
}
```

What is printed when the following code segment is executed in a client class of FrequentFlyer?

```
    FrequentFlyer akshay = new FrequentFlyer(20000);
    FrequentFlyer bruno = new FrequentFlyer(10000);
    FrequentFlyer cindy = new FrequentFlyer(0);
    FrequentFlyer[] friends = {akshay, bruno, cindy};
    int total = 0;
    for (FrequentFlyer p : friends)
    {
        p.addMiles(1000);
        total += p.getMiles();
    }
    System.out.println(total);
```

- (A) 0
- (B) 3000
- (C) 30000
- (D) 33000

34. Consider the following code segment.

```
int[] nums = new int[8];
nums[0] = 0;
int n = 1;

while (n < nums.length)
{
    int k;

    for (k = n; k < 2*n; k++)
    {
        nums[k] = nums[k-n] + 1;
    }

    n = k;
}
```

Which of the following represents the contents of `nums` after the code segment has been executed?

- (A) 0 1 1 1 1 1 1 1
- (B) 0 1 0 1 0 1 0 1
- (C) 0 1 1 2 1 2 2 3
- (D) 0 1 2 3 1 2 3 4

35. Consider the following method.

```
private int mystery(int n)
{
    if (n <= 1)
    {
        return n;
    }
    else
    {
        return n + mystery(n-2);
    }
}
```

What value does `mystery(6)` return?

- (A) 10
- (B) 11
- (C) 12
- (D) 13

36. Consider the following code segment.

```
String[] a = {"A", "B"};
String[] b = a;
String[] c = {a[0], a[1]};

String letter = a[0];
a[0] = a[1];
a[1] = letter;

System.out.println(a[0] + a[1] + b[0] + b[1] + c[0] + c[1]);
```

What is printed when the code segment is executed?

- (A) BAABAB
- (B) BAABBA
- (C) BABAAB
- (D) BABABA

37. Consider the following code segment.

```
ArrayList<Integer> list = new ArrayList<Integer>();
list.add(1);
list.add(2);

for (int i = 1; i <= 3; i++)
{
    list.add(i, i);
}

System.out.println(list);
```

What is the result when the code segment is executed?

- (A) [1, 1, 2, 3, 2] is printed
- (B) [1, 1, 2, 2, 3] is printed
- (C) [1, 2, 1, 2, 3] is printed
- (D) [1, 2, 3, 1, 2] is printed

38. The Binary Search algorithm is designed to work with an array sorted in ascending order. Under certain circumstances the algorithm may find a given target value even if the array is not sorted. For which of the following arrays and target values will standard Binary Search NOT find the target value 5?

- (A) {4, 2, 1, 5, 3, 7, 6}
- (B) {1, 2, 3, 7, 4, 5, 6}
- (C) {1, 2, 3, 4, 5, 6, 7}
- (D) {1, 4, 5, 6, 2, 7, 3}

39. Suppose `m` is a two-dimensional array of size 4 by 4, with all its elements initialized to zeros. The method `fill` is defined as follows:

```
public void fill(int[][] m)
{
    int n = m.length;

    for (int i = 1; i < n - 1; i++)
    {
        for (int j = 1; j < n - 1; j++)
        {
            m[i][j] = 1;
        }
    }
}
```

What values will be stored in `m` after `fill(m)` is called?

- (A) 1100
1100
0000
0000
- (B) 0000
0110
0110
0000
- (C) 1110
1110
1110
0000
- (D) 1111
1111
1111
1111

40. Consider the following class.

```
public class Question
{
    private boolean answer;

    public void flip(Question q)
    {
        < missing statement >
    }

    /* constructors and other methods not shown */
}
```

Which of the following could replace *< missing statement >* in the flip method so that it compiles with no errors?

- (A) `answer = !answer;`
- (B) `answer = !q.answer;`
- (C) `q.answer = !q.answer;`
- (D) All of the above

41. Consider the following code segment.

```
ArrayList<String> xyz = new ArrayList<String>();
xyz.add("X");
xyz.add("Y");
xyz.add("Z");

int count = 0;
for (String s1 : xyz)
{
    for (String s2 : xyz)
    {
        if (s1.equals(s2))
        {
            count++;
        }
    }
}

System.out.print(count);
```

What is the result when the code segment is compiled/executed?

- (A) 0 is printed
- (B) 1 is printed
- (C) 3 is printed
- (D) NullPointerException

42. Kevin has written code to read a text file that contains people's first names, middle names and/or middle initials, and last names. The code is supposed to print the names in a different format: last name, followed by a comma, a space, the first name, and the middle name or initial (if present). Kevin's test file, `Names.txt`, has only three lines:

```
Peter Pan
Wile E. Coyote
John Jacob Jingleheimer Schmidt
```

The expected output is

```
Pan, Peter
Coyote, Wile E.
Schmidt, John Jacob Jingleheimer
```

Here is Kevin's code:

```
Scanner input = new Scanner("Names.txt");

while (input.hasNext())
{
    String[] tokens = input.nextLine().split(" ");
    int n = tokens.length;
    String result = tokens[tokens.length-1] + ",";
    for (int i = 0; i < tokens.length-1; i++)
    {
        result += " " + tokens[i];
    }
    System.out.println(result);
}
input.close();
```

Unfortunately, it has a bug. It turns out that the `Scanner` class has a constructor that takes a string as a parameter. It treats that string as an *input stream*, and all `Scanner` methods that apply to a text file also apply to that input stream in a similar way.

What is the result when Kevin's code is executed?

- (A) `ArrayIndexOutOfBoundsException`
- (B) Only `Names.txt`, is printed
- (C) Only `Pan`, is printed
- (D) `IOException`

Practice Exam #4

SECTION II

Time — 90 minutes

Number of questions — 4

Percent of total grade — 45

1. Many websites have strict requirements for passwords. The class `Password` contains a method that generates a “strong” password of twelve randomly chosen characters according to the rules described below, and another method that generates a password from a secret phrase and a number. You will write these methods in Part (a) and Part (b) of this question, respectively.

A partial definition of the `Password` class is shown below.

```
public class Password
{
    private static final String ABC = "ABCDEFGHIJKLMNOPQRSTUVWXYZ";
    private static final String abc = "abcdefghijklmnopqrstuvwxyz";
    private static final String digits = "0123456789";
    private static final String symbols = "@#!$. ";

    /**
     * Returns a one-character substring of str
     * at a randomly chosen index.
     */
    private String randomChar(String str)
    { /* implementation not shown */ }

    /**
     * Generates and returns a "strong" password, which consists
     * of twelve randomly chosen characters: an uppercase letter,
     * followed by three digits, followed by seven lowercase
     * letters, followed by one symbol. Letters and digits can
     * be repeated.
     */
    public String suggestStrongPassword()
    { /* to be implemented in Part (a) */ }
```

Continued



```

/**
 * Generates and returns a password by taking the first
 * letters of the words in phrase, followed by the digits
 * of n taken in reverse order.
 * Preconditions: phrase consists of several words with
 *                   one space between the words;
 *                   phrase begins and ends with a letter;
 *                   n > 0
 */
public String makePassword(String phrase, int n)
{ /* to be implemented in Part (b) */ }

/* other instance variables, constructors, and methods
   not shown */
}

```

- (a) Write a method `suggestStrongPassword` that generates a random password of 12 characters: an uppercase letter, followed by three digits, followed by seven lowercase letters (letters and digits can be repeated), followed by one symbol. The symbol comes from the `String` `symbols` in the definition of the `Password` class. A sample password may look like this: "L767kyikmuo#".

Complete the method `suggestStrongPassword` below. Do not call any `String` methods in your solution to Part (a).

```

/**
 * Generates and returns a "strong" password, which consists
 * of twelve randomly chosen characters: an uppercase letter,
 * followed by three digits, followed by seven lowercase
 * letters, followed by one symbol. Letters and digits can
 * be repeated.
 */
public String suggestStrongPassword()

```

- (b) Many internet users find it difficult to remember their passwords for various websites. One user decided to take a secret phrase and make a string out of the first letters of the words in that string. `Password`'s method `makePassword` takes a string `phrase` and an integer `n` as parameters, and returns a password made up of the first letters of the words in `phrase` followed by the digits of `n` in reverse order. `phrase` begins and ends with a letter and consists of several words separated by one space.

For example, a call

```
makePassword("Two roads diverged in a yellow wood", 1915)
```

returns "Trdiayw5191".

Complete the method `makePassword` below.

```
/**
 * Generates and returns a password by taking the first
 * letters of the words in phrase, followed by the digits
 * of n taken in reverse order.
 * Preconditions: phrase consists of several words with
 *                   one space between the words;
 *                   phrase begins and ends with a letter;
 *                   n > 0
 */
public String makePassword(String phrase, int n)
```

❖ ❖ ❖

Class information for this question

```
public class Password
{
    private static final String ABC
    private static final String abc
    private static final String digits
    private static final String symbols

    private String randomChar(String str)
    public String suggestStrongPassword()
    public String makePassword(String phrase, int n)
}
```

⤵ For additional practice, add a constructor that takes a string as a parameter and sets symbols to that string. Implement Password's randomChar method.

⤴ Supply a main method in a client of the Password class to test your code.

2. You can purchase any number of apples at Papas Orchards, but every six apples purchased are priced as five. For example, if the price of each apple is 69 cents and you purchase 34 apples, your total cost will be $29 * \$0.69 = \20.01 , because 30 apples are priced as 25. The class `ApplesPurchase` helps calculate the cost of your purchase at Papas.

The constructor of the `ApplesPurchase` class takes two `int` parameters.

- The first parameter is the price of one apple in cents.
- The second parameter is the number of apples purchased.

Assume that the values of both parameters passed to the constructor are positive.

`ApplesPurchase` has a method `totalCost`, which takes no parameters and returns a `String` that includes the quantity of apples purchased, the unit price in cents, and the total cost of the purchase, in the format shown below. For example, if 12 apples are purchased at 50c each, the total cost is \$5.00, and the returned string is

12 apples at 50c each: \$5.00 total

The following table contains a sample code execution sequence and the corresponding values of the string returned by the calls to the `totalCost` method. The code execution sequence appears in a class other than `ApplesPurchase`.

Statements and Expressions	Value Returned (blank if no value)	Comment
<code>ApplesPurchase purchase1 = new ApplesPurchase(79, 2);</code>		<code>ApplesPurchase</code> object <code>purchase1</code> is created.
<code>purchase1.totalCost()</code>	2 apples at 79c each: \$1.58 total	$2 * 79 = 158$
<code>ApplesPurchase purchase2 = new ApplesPurchase(50, 12);</code>		<code>ApplesPurchase</code> object <code>purchase2</code> is created.
<code>purchase2.totalCost()</code>	12 apples at 50c each: \$5.00 total	$2 * 6 \implies 2 * 5$
<code>ApplesPurchase purchase3 = new ApplesPurchase(69, 34);</code>		<code>ApplesPurchase</code> object <code>purchase3</code> is created.
<code>purchase3.totalCost()</code>	34 apples at 69c each: \$20.01 total	1 apple free for every six.

Write a complete `ApplesPurchase` class. Your implementation must meet all specifications, and the format of the output must conform to the examples shown in the table.

¶ For additional practice, derive a class `ApplesPurchaseWithPie` from `ApplesPurchase` to provide a way to add an apple pie to the purchase.

Apple pies should be available in two sizes, small ("S") and large ("L"). Pass the price of the large pie as a parameter to `ApplesPurchaseWithPie`'s constructor. Compute the price of the small pie as 75% of the large pie price, rounded to the nearest cent.

¶ Override `ApplesPurchase`'s `totalCost` method.

3. In an international ballroom dancing competition, competitors are scored by a panel of judges on a scale from 0 to 100. Each couple's event score is computed as the average of the scores from all the judges (possibly excluding the highest score, as explained below).

The competition's scoring is modeled by the `BallroomCompetition` class. A partial definition of this class is shown below. You will write one method of this class.

```
public class BallroomCompetition
{
    /**
     * Returns the number of occurrences of target in scores.
     */
    private int countOccurrences(
        ArrayList<Integer> scores, int target)
    { /* implementation not shown */ }

    /**
     * Returns the average of the values in scores. However,
     * if the size of the scores list is 6 or greater and
     * the largest value occurs only once in scores, that value
     * is excluded from computing the average.
     * Preconditions: scores.size() >= 3;
     *                  0 <= scores[k] <= 100
     */
    public double averageScore(ArrayList<Integer> scores)
    { /* to be implemented */ }

    /* other instance variables, constructors, and methods
       not shown */
}
```

Write a method `averageScore` of the `BallroomCompetition` class. It takes an `ArrayList<Integer> scores` as a parameter and computes and returns the average score. However, if the size of the list `scores` is 6 or greater and the maximum value occurs only once in the list, that value is excluded from computing the average.

The `BallroomCompetition` class has a private method

```
private static int countOccurrences(
    ArrayList<Integer> scores, int target)
```

that returns the number of occurrences of `target` in `scores`. In writing `averageScore`, call the `countOccurrences` method.

Complete the method `averageScore` below.

```
/**
 * Returns the average of the values in scores. However,
 * if the size of the scores list is 6 or greater and
 * the largest value occurs only once in scores, that value
 * is excluded from computing the average.
 * Preconditions: scores.size() >= 3;
 *                  75 <= scores[k] <= 100
 */
public double averageScore(ArrayList<Integer> scores)
```

❖ ❖ ❖

Class information for this question

```
public class BallroomCompetition
private int countOccurrences(ArrayList<Integer> scores,
                             int target)
public double averageScore(ArrayList<Integer> scores)
```

 For additional practice, implement the `countOccurrences` method.

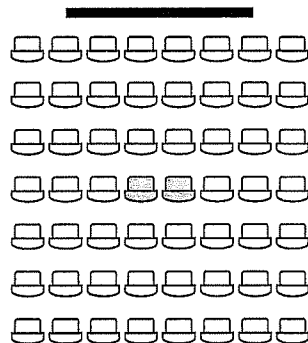
Implement a computer model for a ballroom dancing competition. The competition includes several dance events, and several couples participate in all the events. Each event is scored by a panel of eight judges. Generate the judges' scores as random integers from 75 to 100. Calculate the average score for each couple in each event by calling the `averageScore` method. Multiply the average by the total number of judges, round the result to the nearest integer, and add it to the couple's final cumulative score. For three events and eight judges, the cumulative score for each couple should be between $75 \cdot 8 \cdot 3 = 1800$ and $100 \cdot 8 \cdot 3 = 2400$.

Determine the maximum total score among all the couples, and find the number of couples that share first place.

 Whenever possible, use the appropriate methods of the `BallroomCompetition` class — do not duplicate their code.

4. The seats in a movie theater are modeled by a two-dimensional array of boolean values: `true` represents an occupied seat, and `false` represents a vacant seat. `seats[0]` represents the row of seats closest to the screen. The seats in the theater are arranged in a rectangle and the number of seats in each row is even.

Let's say that the optimal pair of seats is in the center of the fourth row, as shown in the picture below.



The theater is modeled by the class `MovieTheater`. A partial definition of this class is shown below.

```
public class MovieTheater
{
    /** The seats in the theater. true indicates an occupied seat;
        false indicates a vacant seat. */
    private boolean[][] seats;

    /**
     * Initializes the instance variable seats to a 2D array with
     * the same dimensions as the dimensions of the given
     * 2D array m. Each element of seats is set to true if the
     * corresponding element in m is 1; otherwise it is set
     * to false.
     * Preconditions: m.length >= 6;
     *                     the number of columns is even
     */
    public MovieTheater(int[][] m)
    { /* implementation not shown */ }
```

Continued



```
/**
 * Returns the location of the pair of adjacent vacant seats
 * that is closest to the optimal pair of seats, excluding
 * the first two rows. The location of the left seat
 * (smaller column index) in the pair is used.
 * If several locations are at the same distance,
 * returns any one of them.
 * If no such pair of seats is found, returns null.
 */
public Location findBestPairOfSeats()
{ /* to be implemented */ }

/* other instance variables, constructors, and methods
not shown */
}
```

The `MovieTheater` class uses the class `Location`. A partial definition of `Location` is shown below. `Location` has a constructor that takes two integer parameters — row and column. It also has an `int` method `getDistance(Location other)`, which returns the distance from this location to `other`.

```
public class Location
{
    public Location(int r, int c)
    { /* implementation not shown */ }

    /**
     * Returns the distance from this location to other.
     */
    public int getDistance(Location other)
    { /* implementation not shown */ }

    /* instance variables and other methods not shown */
}
```

You will write the method `findBestPairOfSeats` of the `MovieTheater` class. This method returns the `Location` of the pair of adjacent vacant seats closest to the optimal pair of seats in the middle of the fourth row, but not in the first two rows. The location of the left-hand seat (smaller column index) in the found pair is returned. If several such locations are at the same best distance, the method returns any one of them. If no such pairs of adjacent vacant seats are found, `findBestPairOfSeats` returns `null`.

Complete the `findBestPairOfSeats` method below.

```
/**
 * Returns the location of the pair of adjacent vacant
 * seats closest to the optimal location, excluding
 * the first two rows. The location of the left seat
 * (smaller column index) in the pair is used.
 * If several locations are at the same distance,
 * returns any one of them.
 * If no such pair of seats is found, returns null.
 */
public Location findBestPairOfSeats()
```

❖ ❖ ❖

Class information for this question

```
public class Location
public Location(int r, int c)
public int getDistance(Location other)

public class MovieTheater
private boolean[][] seats

public MovieTheater(int[][] m)
public Location findBestPairOfSeats()
```

¶ For additional practice, implement the `Location` class. Make the distance between seats equal to the absolute value of the difference in rows plus the absolute value of the difference in columns. Also add a reasonable `toString` method.

¶ Write the constructor for the `MovieTheater` class and test both classes.